

SHIVANSH SHARMA

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PROFESSIONAL SUMMARY

Backend engineer with 3+ years building high-throughput Java/Kafka distributed systems. Currently on Oracle's Clinical Data Exchange Hub (CDeX), moving clinical documents between healthcare providers and payers, handling 12M+ weekly events across 55M+ patient records. At ZL Technologies, cut 50M-record re-indexing from 24h to 2h.

PROFESSIONAL EXPERIENCE

Software Development Engineer 2

Jun 2025 – Present

Oracle Remote

- **Co-designed a 3-microservice document-search subsystem**, authoring design RFCs and data flow, building one service and co-developing the rest; handling 25% of CDeX's 12M+ weekly events, cutting peak CPU from 90% to 65%.
- **Cut document transmission failures by 25%** across 500K+ weekly exchanges by reworking retry logic with exponential backoff, jitter, and idempotency checks, and closing edge cases in collaboration with QA.
- **Built Docker-based MinIO and Oracle DB simulators** from scratch, giving the team a consistent local test environment and cutting pre-deployment testing time by 30%.
- **Raised unit test coverage from 60% to 85%** by introducing AI-assisted PR reviews to flag untested code paths and adding JaCoCo coverage checks to the Maven build.
- **Wrote canary test scenarios for 5 production services** to verify deployment sanity, and set up REST-based inter-service communication across the services, enabling faster rollback when a deployment fails.

Software Development Engineer

Jul 2023 – Jun 2025

ZL Technologies Hyderabad, India

- **Led a 4-engineer team as tech lead:** broke PRD requirements into sprint tasks, assigned work to junior engineers, and reviewed the juniors' code to deliver on time and to spec.
- **Raised archiving accuracy from 60% to 80%** (records archived without loss or duplication) by building distributed archiving on message queues, fault-tolerant MapReduce, and JDBC batching.
- **Slashed 50M-record re-indexing time by 90%** (24h to 2h) using a Lucene "Index of Index" architecture, resolving analytics bottlenecks and restoring cross-team SLAs.
- **Migrated from a proprietary tool to Apache Tika** per the senior leads' design, building the integration and benchmarking extraction quality against the old tool to eliminate \$1M in annual licensing fees.

TECHNICAL SKILLS

- **Core Competencies:** Data Structures & Algorithms (DSA), Distributed Systems, System Design (HLD/LLD), Microservices, Event-Driven Architecture
- **Languages & Frameworks:** Java (primary), SQL, C++ (DSA), Spring Boot, Micronaut
- **Messaging & Search:** Kafka, Lucene, Apache Tika, MapReduce
- **Cloud & Infrastructure:** OCI (OKE), AWS (EC2, S3, IAM, VPC), Docker, Kubernetes, Terraform, MinIO
- **Security:** JWT (service authentication), OAuth2, TLS certificate management (DigiCert)
- **Databases & Caching:** Oracle ADB, MySQL, PostgreSQL, MongoDB, Redis
- **Monitoring & Logging:** Splunk, New Relic, Grafana, Prometheus, OCI Monitoring Query Language (MQL)
- **Testing & Tools:** JUnit, Mockito, JaCoCo, Git, Maven

EDUCATION

Indian Institute of Technology Roorkee (IIT Roorkee)

Jul 2019 – May 2023

Bachelor of Technology, Civil Engineering

Roorkee, India

KEY PROJECTS

Scalable Discussion Forum Platform | Java, Spring Boot, Kafka, MySQL, Redis

- Built an event-driven forum on Kafka with idempotent write APIs (request keys and transactions), rate limiting, and stateless load balancing, load-tested to 10K+ concurrent users and 5K+ RPS.
- Sustained 99.9% read availability while killing containers under load, measured by successful reads, applying CAP trade-offs in schema design.